Lectures Review: Nonparametric Prediction & Tree Ensembles

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Outline

1. Nearest Neighbors and Metric Geometry
2. Decision Trees and Adaptive Partitioning
3. Ensemble Learning and Random Forests
4. Summary

Problem setup: local estimation from spatial proximity

How can we predict target y for a query x using only the spatial proximity of historical data, and when does Euclidean metric fail?

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How can we predict target y for a query x using only the spatial proximity of historical data, and when does Euclidean metric fail?

Given Data: Historical training pairs $\mathcal{D}_N = \{(x_1, y_1), \ldots, (x_N, y_N)\} \subset \mathbb{R}^d \times \mathcal{Y}$.

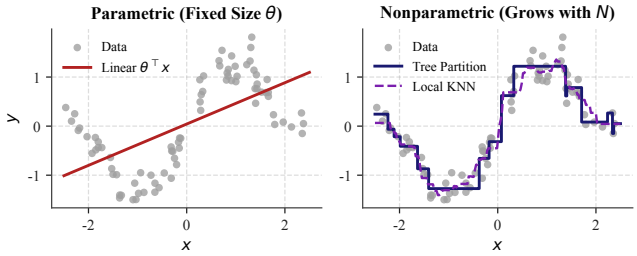
Metric Smoothness: Points close in input space share similar conditional label distributions:

$$x \approx x' \Longrightarrow \mathbb{P}(Y \mid X = x) \approx \mathbb{P}(Y \mid X = x').$$

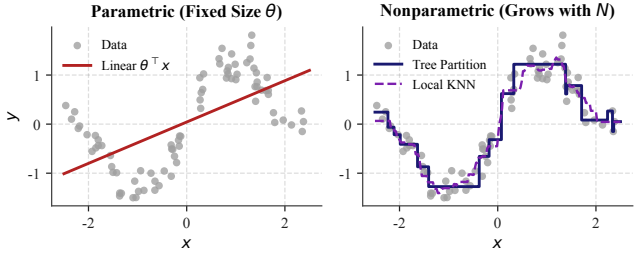
Nonparametric Goal: Predict $\hat{y}(x)$ locally without imposing rigid parametric forms.

Central Dilemma: In high dimensions (d**g**1), does Euclidean "nearness" still preserve similarity?

The foundation: parametric versus nonparametric models



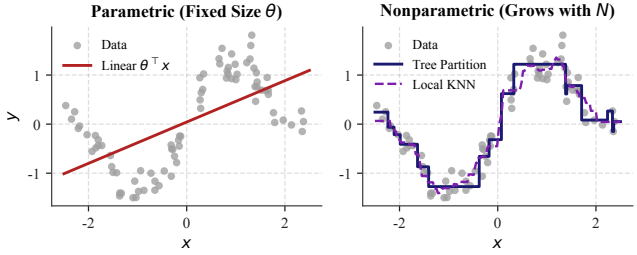
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Parametric vs Nonparametric Learning.

- **Parametric:** Fixed parameter vector $\boldsymbol{\theta} \in \mathbb{R}^p$; fixed capacity risks model misspecification.

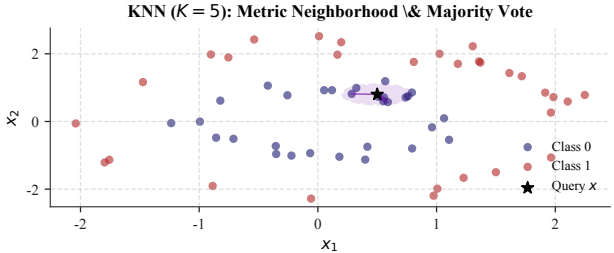
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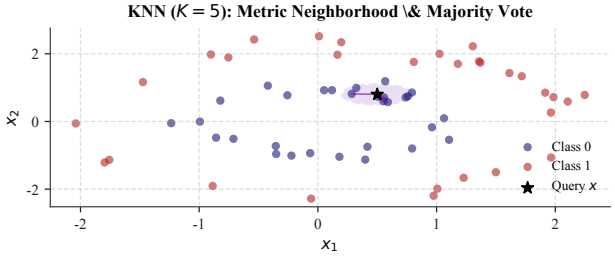
Parametric vs Nonparametric Learning.

- **Parametric:** Fixed parameter vector $\boldsymbol{\theta} \in \mathbb{R}^p$; fixed capacity risks model misspecification.
- **Nonparametric:** Hypothesis space $\mathcal{F}_N$ expands with data N, trading variance for zero bias.

K-nearest neighbors: metric and voting rules



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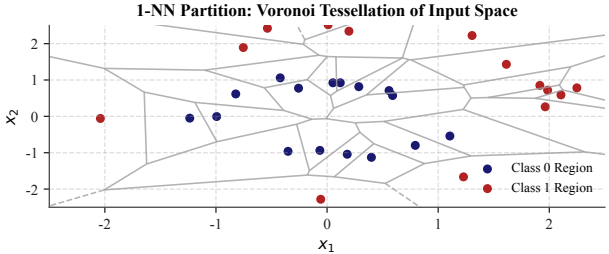


Decision Rules on Neighborhood $\mathcal{N}_K(x)$.

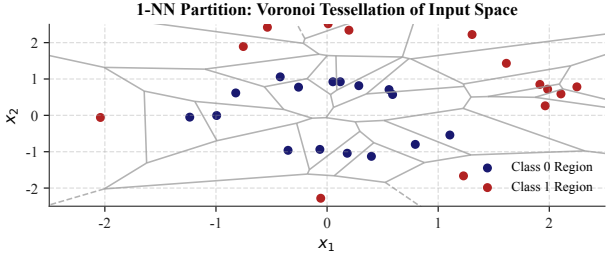
$$\text{Classification (vote): } \hat{y}(x) = \arg\max_c \sum_{i \in \mathcal{N}_K(x)} \mathbf{1}\{y_i = c\}, \quad \text{Regression (mean): } \hat{y}(x) = \frac{1}{K} \sum_{i \in \mathcal{N}_K(x)} y_i.$$

K-NN is a memory-based lazy learner: data is stored; all computation is deferred to query time.

Geometry of 1-NN: Voronoi tessellation of space



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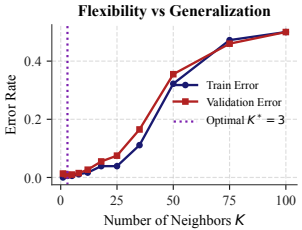


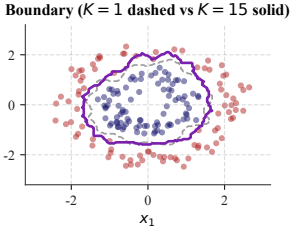
Voronoi Cell Partition. Every training point x_i defines a convex polygonal cell:

$$V_i := \{x \in \mathcal{X} : \|x - x_i\|_2 \leq \|x - x_j\|_2, \quad \forall j \neq i\}.$$

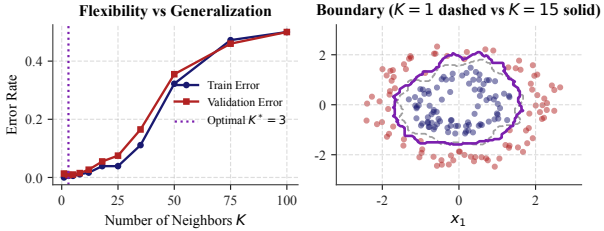
Decision boundaries are formed by perpendicular bisectors between neighboring training points.

Choosing K: flexibility, bias, and variance



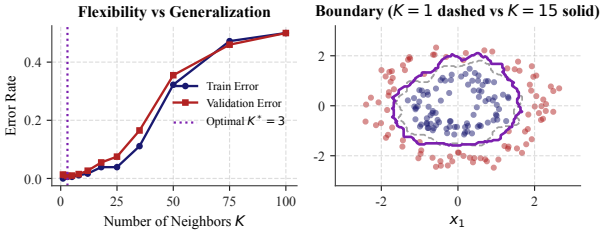


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Small K ($K = 1$): High flexibility, zero training error, low bias, but high variance (noise sensitive).

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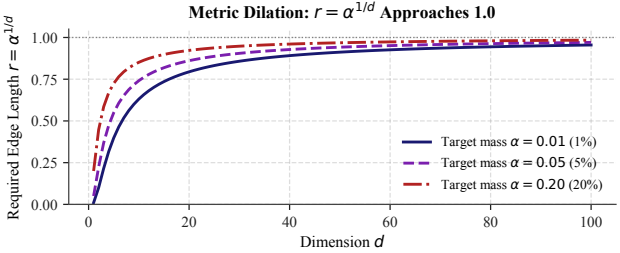


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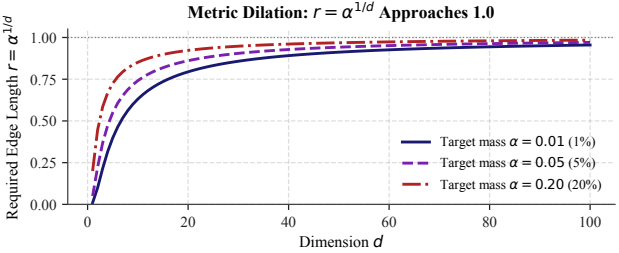
Large K ($K \to N$): Heavy smoothing, low variance, but high bias (misses local boundaries).

Optimal K^* is selected at the minimum of the validation error curve.

Curse of dimensionality: volume dilation of subcubes



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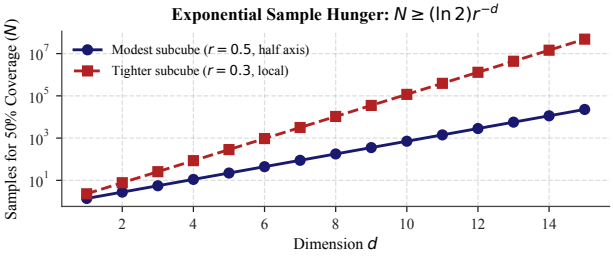


Volume Scaling on Uniform $[0, 1]^d$**.** Target fraction α = Volume = $r^d \implies \mathbf{r = \alpha^{1/d}}$.

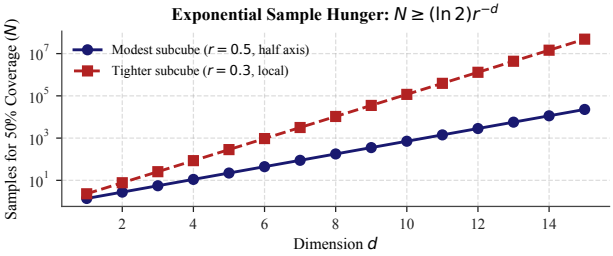
To capture $\alpha = 0.01$ (1% volume): $d = 1 \implies r = 0.01$; $d = 10 \implies r \approx 0.63$; $d = 100 \implies r \approx 0.955$.

In high dimensions, neighborhoods must span almost the entire axis; locality completely breaks down.

Curse of dimensionality: exponential sample hunger



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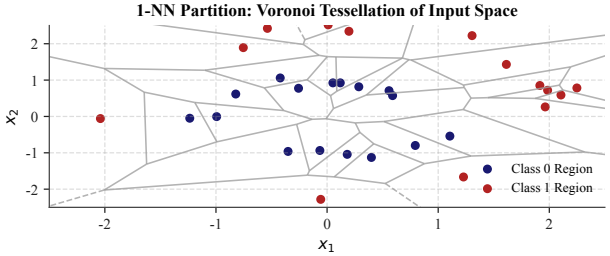


Coverage Requirement for Contained Subcube of Side r.

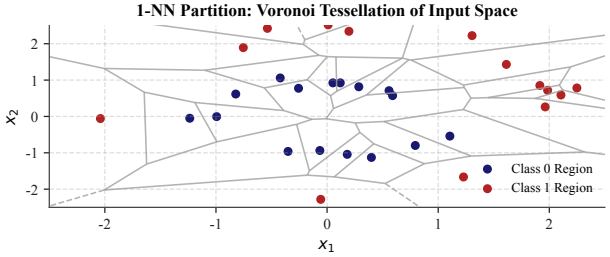
$$\mathbb{P}(\text{at least one sample inside}) = 1 - (1 - r^d)^N \geq \frac{1}{2} \implies \mathbf{N} \geq \frac{\ln \mathbf{2}}{-\ln(\mathbf{1} - \mathbf{r^d})} \approx (\ln \mathbf{2})\mathbf{r^{-d}}.$$

Maintaining local density requires sample size scaling exponentially with dimension: $N \in \Omega((1/r)^d)$.

Asymptotic guarantee: the Cover-Hart bound



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Cover-Hart Theorem (1967). For Bayes risk $R^* = \mathbb{E}[\min(\eta(X), 1 - \eta(X))]$, the asymptotic 1-NN risk satisfies:

$$\lim_{N\to\infty} R(f_{1\text{NN}}) = 2\,\mathbb{E}\big[\eta(X)(1-\eta(X))\big] \le 2R^*(1-R^*) \le \mathbf{2R}^*.$$

Without knowing true densities, 1-NN error is asymptotically at most twice the optimal Bayes error!

Section 1 Takeaways: Geometry, KNN, and the Curse of Dimension

Core Principles to Remember:

- **Nonparametric Nature:** No rigid functional forms; 1-NN error is bounded by twice the Bayes risk: $R^* \leq R_{\text{1-NN}} \leq 2R^*$.
- **Bias-Variance Trade-off:** K controls smoothing: $K = 1$ has low bias but high variance; large $K \to N$ reduces variance at the expense of high boundary bias.
- **Curse of Dimensionality:** Volume scaling $r = \alpha^{1/d} \to 1.0$ forces neighborhoods to span the full space; required samples scale as $N \in \Omega((1/r)^d)$.
- **Metric Isotropy Dilemma:** Euclidean distance treats all coordinates equally, diluting meaningful signals in the presence of noise features.

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Problem setup: label-guided space partitioning

Instead of isotropic distance balls, can we let target labels guide an adaptive, axis-aligned orthogonal partition that selects informative features and ignores noise?

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Instead of isotropic distance balls, can we let target labels guide an adaptive, axis-aligned orthogonal partition that selects informative features and ignores noise?

Given Data: Dataset $\mathcal{D}_N = \{(x_i, y_i)\}_{i=1}^N \subset \mathbb{R}^d \times \mathcal{Y}$ with mixed feature scales and noise coordinates.

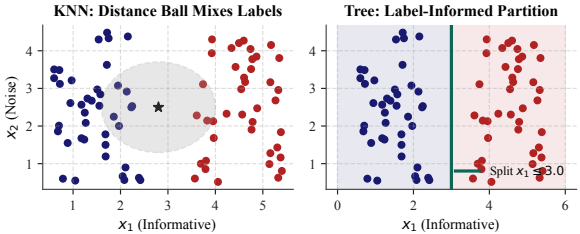
Orthogonal Partitioning: Slices the feature space into M disjoint hyperrectangles $R_1, \ldots, R_M$:

$$f(x) = \sum_{m=1}^{M} c_m \mathbf{1}\{x \in R_m\}, \quad \bigcup_{m=1}^{M} R_m = \mathcal{X}, \quad R_m \cap R_{m'} = \emptyset.$$

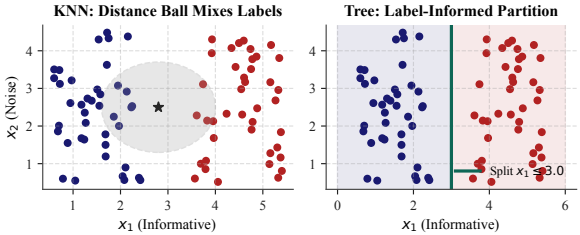
Goal: Maximize label purity in each sub-region with simple axis-aligned threshold tests.

Central Dilemma: Finding optimal trees is NP-complete; how do greedy heuristic splits behave?

Label-guided regions: overcoming metric isotropy

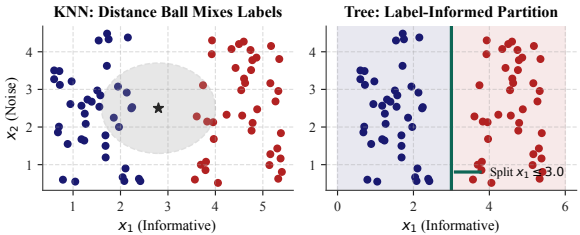


Label-guided regions: overcoming metric isotropy



KNN (Isotropic Metric): Computes distance across all coordinates, mixing noisy features.

Label-guided regions: overcoming metric isotropy

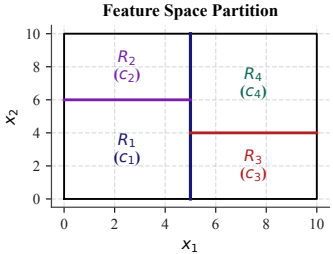


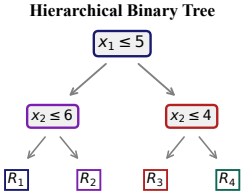
KNN (Isotropic Metric): Computes distance across all coordinates, mixing noisy features.

Decision Tree (Label-Guided): Splits on informative feature x_1 and ignores noisy x_2.

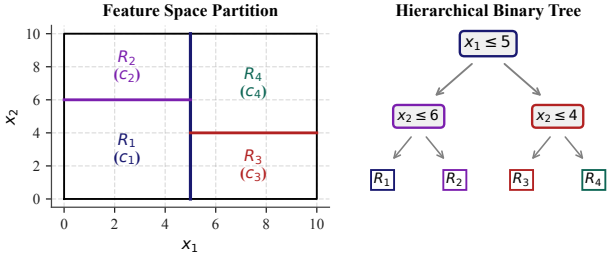
Decision trees replace isotropic distance balls with label-guided axis-aligned splits.

Recursive partitioning: step-functions and hierarchy





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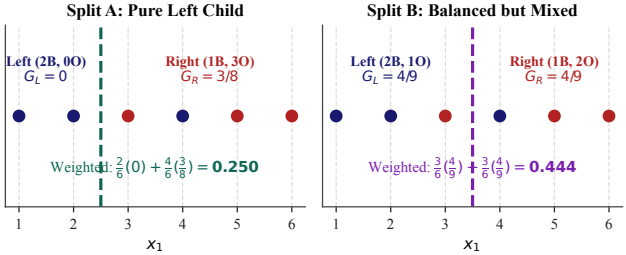


Piecewise Constant Step-Function Representation.

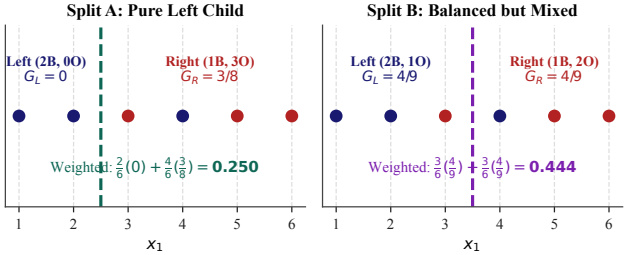
$$f(x) = \sum_{m=1}^{M} c_m \mathbf{1}\{x \in R_m\}, \qquad \bigcup_{m=1}^{M} R_m = \mathcal{X}, \quad R_m \cap R_{m'} = \emptyset.$$

Internal nodes test axis-aligned questions; leaves predict mean $\hat{c}_m$ or majority class.

Scoring splits: Gini impurity and weighted gain



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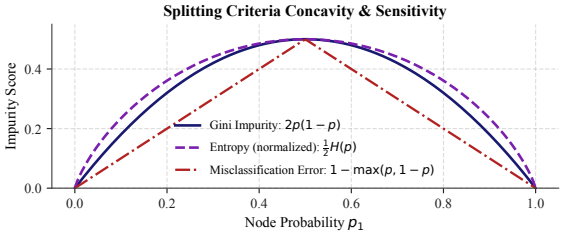


Gini Impurity & Weighted Child Score.

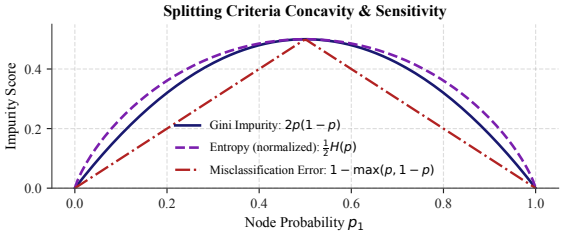
$$G(\mathbf{p}) := \sum_{k=1}^{K} p_k(1 - p_k) = 1 - \sum_{k=1}^{K} p_k^2, \qquad \text{Weighted Score} := \frac{N_L}{N} G(\mathbf{p}_L) + \frac{N_R}{N} G(\mathbf{p}_R).$$

$G(\mathbf{p})$ is the probability that two random draws disagree; CART minimizes weighted child impurity.

Splitting criteria: Gini, entropy, and MSE

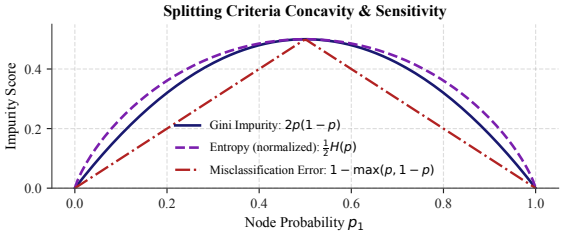


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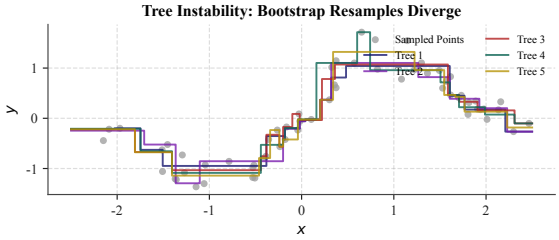
Classification: Gini $1 - \sum p_k^2$, Cross-Entropy $-\sum p_k \log_2 p_k$ (strict concavity ensures $\Delta I \geq 0$).

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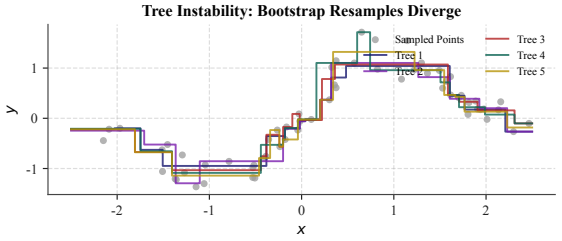


Classification: Gini $1 - \sum p_k^2$, Cross-Entropy $-\sum p_k \log_2 p_k$ (strict concavity ensures $\Delta I \geq 0$).
Regression: Variance / Mean Squared Error MSE $= \frac{1}{N}\sum (y_i - \bar{y})^2$.
Strict concavity guarantees that every split strictly reduces or preserves parent impurity.

Properties and fatal flaw: high structural variance

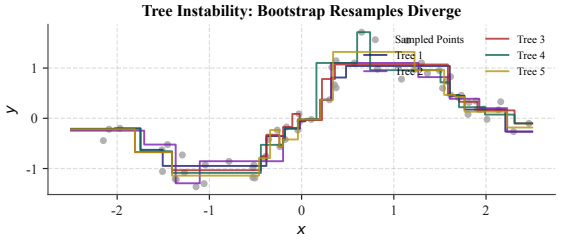


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Advantages: White-box interpretability; strictly invariant to monotonic feature scaling.

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Fatal Flaw: High Variance. Greedy decisions are fragile; tiny data shifts alter root splits! Single trees suffer from severe structural instability $\Longrightarrow$ motivates ensemble learning!

Section 2 Takeaways: Adaptive Partitioning and Single Trees

Core Principles to Remember:

- **Step-Function Model:** Space is cut into orthogonal hyperrectangles R_m; leaves predict local mean $\bar{y}_m$ or majority class.
- **Label-Guided Coordinate Cuts:** Overcomes Euclidean metric isotropy by selecting informative features and discarding irrelevant noise dimensions.
- **Concavity Principle:** Strictly concave impurity criteria (Gini, Entropy) guarantee impurity reduction ($\Delta I \geq 0$ via Jensen's inequality), avoiding flat plateaus of 0-1 error.
- **Scale Invariance & White-Box Logic:** Invariant to strictly monotonic 1D transformations; produces intuitive human-readable decision rules.
- **High Structural Variance:** Greedy top-down splitting propagates root errors downward; small data perturbations cause cascading structural divergence.

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Problem setup: conquering high structural variance

How can we combine multiple unstable, high-variance decision trees into an ensemble that achieves dramatic variance reduction without sacrificing low bias?

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How can we combine multiple unstable, high-variance decision trees into an ensemble that achieves dramatic variance reduction without sacrificing low bias?

Given Estimator: Deep, unpruned decision trees with low bias but notorious structural instability.
Averaging Principle: Averaging B independent estimators scales down variance by $1/B$:

$$\mathrm{Var}\left(\frac{1}{B}\sum_{b=1}^{B} h_b(x)\right) \ll \mathrm{Var}(h(x)).$$

Ensemble Goal: Build an aggregated predictor from a *single* observed dataset S_N.
Central Dilemma: With only one dataset, how do we generate diverse (uncorrelated) trees?

Step-by-step derivation: ensemble variance decomposition

Let $h_1(x), \ldots, h_B(x)$ have variance σ^2 and correlation $\mathrm{Corr}(h_b, h_{b'}) = \rho$ for $b \neq b'$ ($\mathrm{Cov} = \rho\sigma^2$).

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1. Expand Variance of Sample Average $\bar{h}(x) = \frac{1}{B}\sum_{b=1}^{B} h_b(x)$:

$$\mathrm{Var}(\bar{h}(x)) = \frac{1}{B^2}\left[\sum_{b=1}^{B} \mathrm{Var}(h_b(x)) + \sum_{b \neq b'} \mathrm{Cov}(h_b(x), h_{b'}(x))\right]$$

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2. Count B Diagonal and $B(B-1)$ Covariance Terms:

$$= \frac{1}{B^2}\left[B\sigma^2 + B(B-1)\rho\sigma^2\right] = \frac{\sigma^2}{B} + \left(1 - \frac{1}{B}\right)\rho\sigma^2 = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$$

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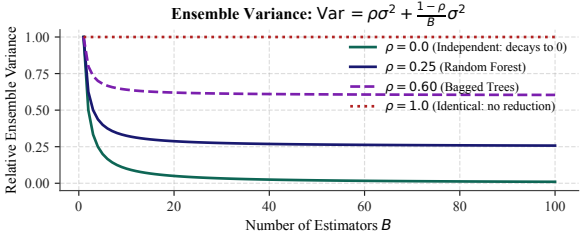
$$\mathrm{Var}(\bar{h}(x)) = \frac{1}{B^2}\left[\sum_{b=1}^{B} \mathrm{Var}(h_b(x)) + \sum_{b \neq b'} \mathrm{Cov}(h_b(x), h_{b'}(x))\right]$$

2. Count B Diagonal and $B(B-1)$ Covariance Terms:

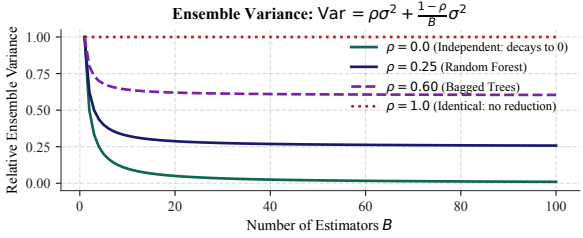
$$= \frac{1}{B^2}\left[B\sigma^2 + B(B-1)\rho\sigma^2\right] = \frac{\sigma^2}{B} + \left(1 - \frac{1}{B}\right)\rho\sigma^2 = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$$

- $\rho\sigma^2$: **Irreducible variance floor** (governed solely by tree correlation ρ).
- $\frac{1-\rho}{B}\sigma^2$: **Reducible variance** (shrinks to 0 as ensemble size $B \to \infty$).

The core math: variance reduction through averaging



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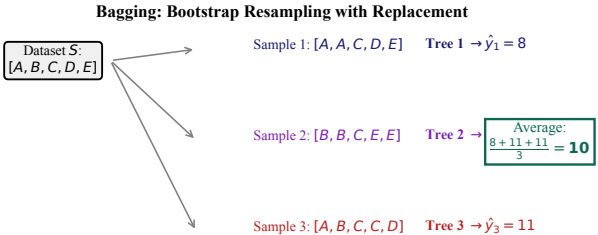


Ensemble Variance Decomposition.

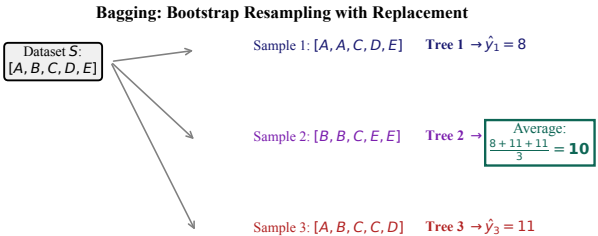
$$\mathrm{Var}\left(\frac{1}{B}\sum_{b=1}^{B} h_b(x)\right) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2.$$

Adding trees only shrinks the second term; to lower the irreducible floor $\rho\sigma^2$, we must suppress ρ!

Bagging: bootstrap aggregation by hand



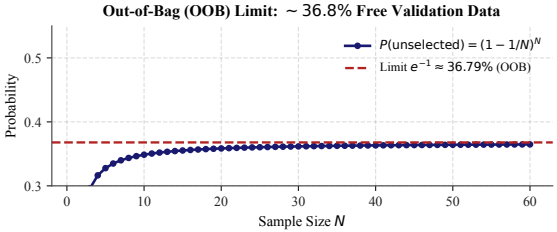
Bagging: bootstrap aggregation by hand



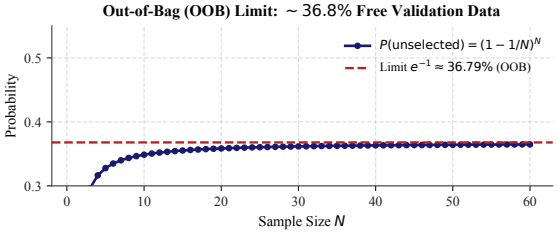
Bootstrap Resampling Protocol. Sample N items with replacement from S_N; train an unpruned deep tree $h_b(x)$ on each multiset.

Predict by simple average $\hat{y} = \frac{1}{B} \sum h_b(x)$ or majority vote. Reuses data to simulate independent samples, reducing variance without requiring new measurements.

Out-of-bag validation: free internal evaluation



Out-of-bag validation: free internal evaluation

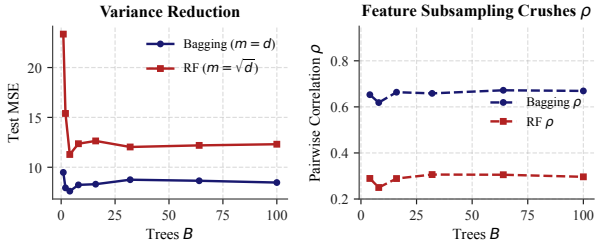


Out-of-Bag (OOB) Limit Probability.

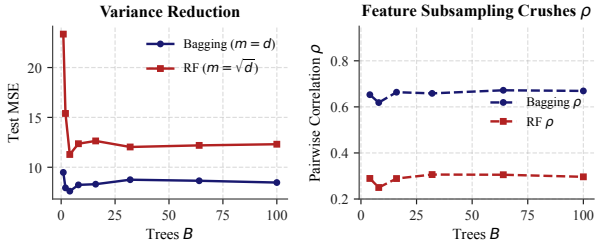
$$p_{\text{OOB}} = \left(1 - \frac{1}{N}\right)^N \xrightarrow{N\to\infty} \frac{1}{e} \approx \mathbf{36.8\%}.$$

Predict on sample i using only trees that omitted i in their bootstrap set. OOB error provides an honest, built-in validation metric without reserving an explicit holdout set!

Random forests: crushing tree correlation via subsampling

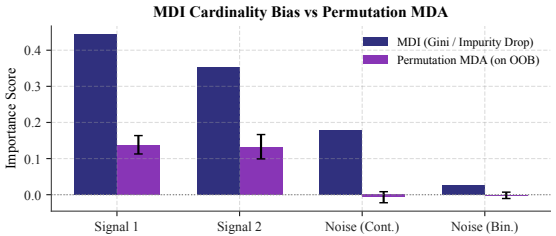


Random forests: crushing tree correlation via subsampling

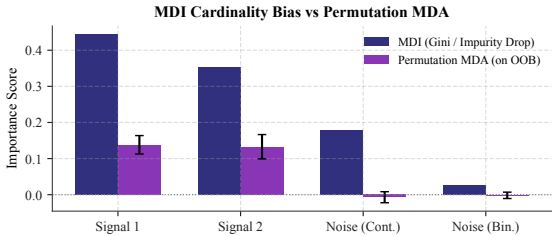


Random Feature Subspace Sampling (Breiman, 2001). At each node split, randomly sample $m \approx \sqrt{d}$ candidate features. Forces trees to explore diverse features, suppressing correlation ρ and sinking the variance floor $\rho\sigma^2$. Massive decorrelation (ρ drops from 0.78 to 0.42) yields superior accuracy!

Feature importance: MDI bias versus permutation honesty

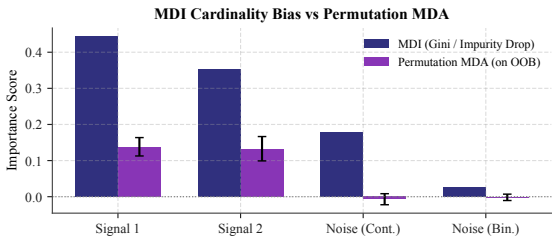


Feature importance: MDI bias versus permutation honesty



MDI (Gini / Impurity Drop): Biased towards high-cardinality continuous features (even pure noise!).

Feature importance: MDI bias versus permutation honesty



MDI (Gini / Impurity Drop): Biased towards high-cardinality continuous features (even pure noise!).
Permutation MDA (on OOB data): Shuffles feature column j; unbiased and rates noise near zero.
Permutation importance breaks the feature-label link, ensuring honest feature attribution.

Outline

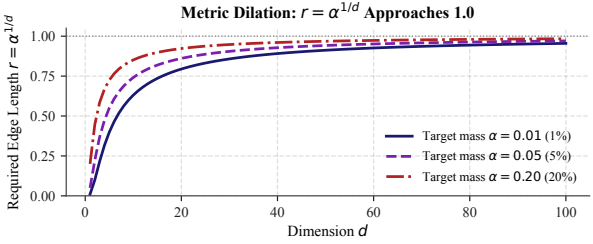
1 Nearest Neighbors and Metric Geometry

2 Decision Trees and Adaptive Partitioning

3 Ensemble Learning and Random Forests

4 Summary

Q1: How Does Dimensionality Affect Local Estimation? I

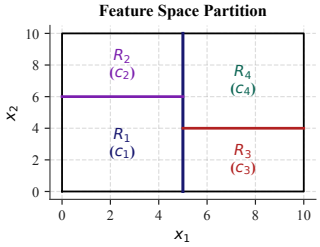


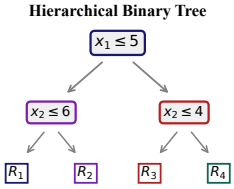
Q1: How Does Dimensionality Affect Local Estimation? II

Exam-Grade Revision Card: Dimensionality vs Locality

- **Metric Dilation:** In $[0,1]^d$, capturing fraction α requires edge $r = \alpha^{1/d} \to 1.0$; neighborhoods lose all locality.
- **Exponential Sample Hunger:** Maintaining fixed local resolution requires $N \in \Omega((1/r)^d)$ samples.
- **Finite vs Asymptotic:** While Cover-Hart bounds 1-NN risk by $2R^*$ as $N \to \infty$, finite-sample convergence rates deteriorate exponentially with d.

Q2: Why Are Decision Trees Adaptively Powerful Yet Fragile? I



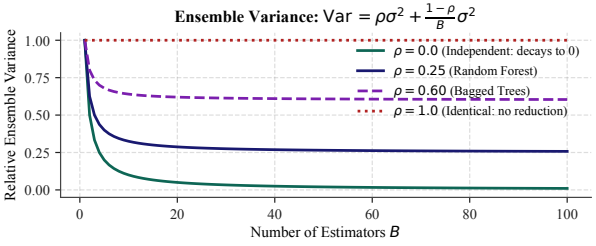


Q2: Why Are Decision Trees Adaptively Powerful Yet Fragile? II

Exam-Grade Revision Card: Adaptive Splits vs Instability

- **Adaptive Power:** Slices single coordinates using strictly concave criteria ($\Delta I \geq 0$); naturally filters out uninformative noise features and is strictly invariant to monotonic scaling.
- **Structural Fragility:** Greedy top-down partitioning cannot backtrack; tiny perturbations near the root cause catastrophic error cascades and structural divergence.
- **The Diagnosis:** Trees are low-bias, high-variance learners demanding variance-reduction ensembles.

Q3: What Is the Exact Variance Mechanism of Random Forests? I

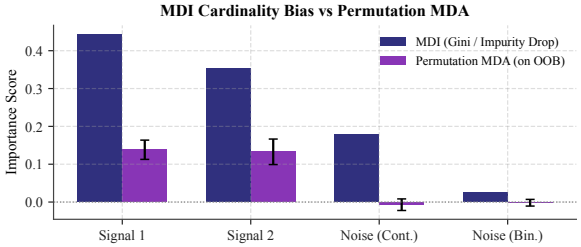


Q3: What Is the Exact Variance Mechanism of Random Forests? II

Exam-Grade Revision Card: Ensemble Variance Reduction

- **Two-Part Formula:** $\mathrm{Var}(\bar{h}) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$. Bagging only eliminates the second term, hitting the $\rho\sigma^2$ floor.
- **Feature Subsampling:** Sampling $m \approx \sqrt{d}$ candidates per split prevents strong features from dominating every root, drastically depressing correlation ρ (e.g., $0.78 \to 0.42$).
- **OOB Bonus:** Resampling with replacement automatically leaves $(1 - 1/N)^N \to 36.8\%$ samples for free internal validation.

Q4: How Should We Interpret Complex Tree Ensembles? I



Q4: How Should We Interpret Complex Tree Ensembles? II

Exam-Grade Revision Card: Honest Feature Attribution

- **MDI Bias Trap:** In-bag impurity drops heavily favor continuous/high-cardinality noise features due to artificial split opportunities.
- **Permutation MDA Standard:** Shuffling feature column j on OOB samples destroys true target dependence; purely uninformative noise drops to zero importance.
- **Best Practice:** Use MDI only for quick screening; rely on OOB Permutation MDA for honest, reliable scientific attribution.